LUNAR BUILD MISSION PLAN — t
Moon · 2026-06-03 · cut-fill balanced, optimized sequence

#	Trip	kind	Site (x,y)	Mass t	Duration	Energy (chg)
1	cut → fill	cutfill	(60,0)	1.66	2.3 d	1.8
2	Excavate spoil: cut	dig	(60,0)	1.79	44.5 h	1.6

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MATERIAL BALANCE
  cut 3.5 t → fill 1.7 t  ·  surplus(spoil) 1.8 t  ·  deficit(import) 0.0 t  ·  sinter 0.00 t
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TOTALS   project 4.2 d (100 h)   moved 3.5 t (115 drum loads)
         energy 16.2 MJ (3.4 charges, 3 recharge stops)   drive 13.92 km
         56 drum cycles; fill SENSED from motor current +/-2.6% (>half full), no load cell
         per-sortie range 28 km (slope+slip @ 17 deg; 32 km flat)
         1 lunar day ~ 30 Earth-days (354 h light)
         ConOps: 70 km + 5-10 t / 11 d -> drive ~2.0 packs vs dig ~4-9 packs: drums dominate
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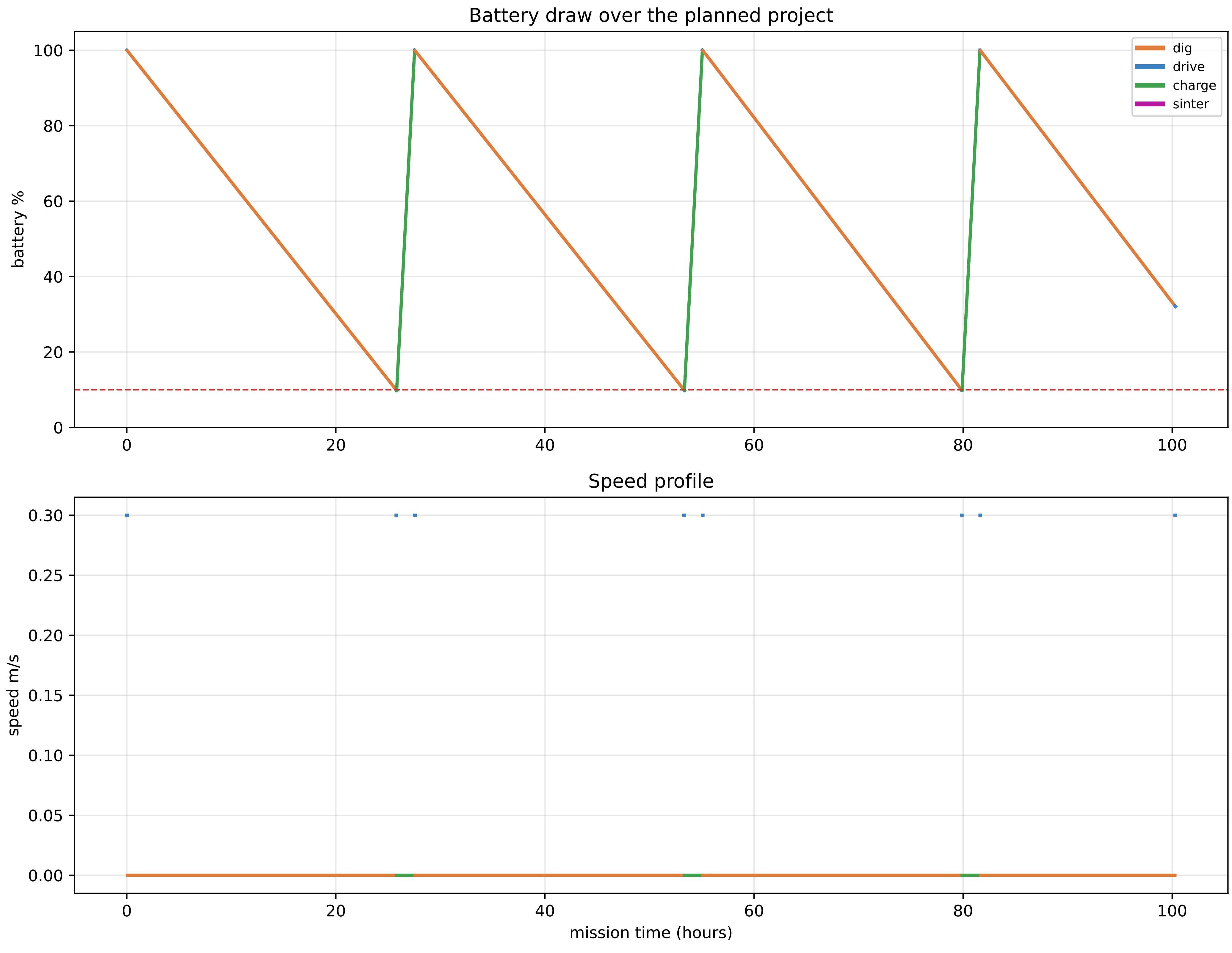
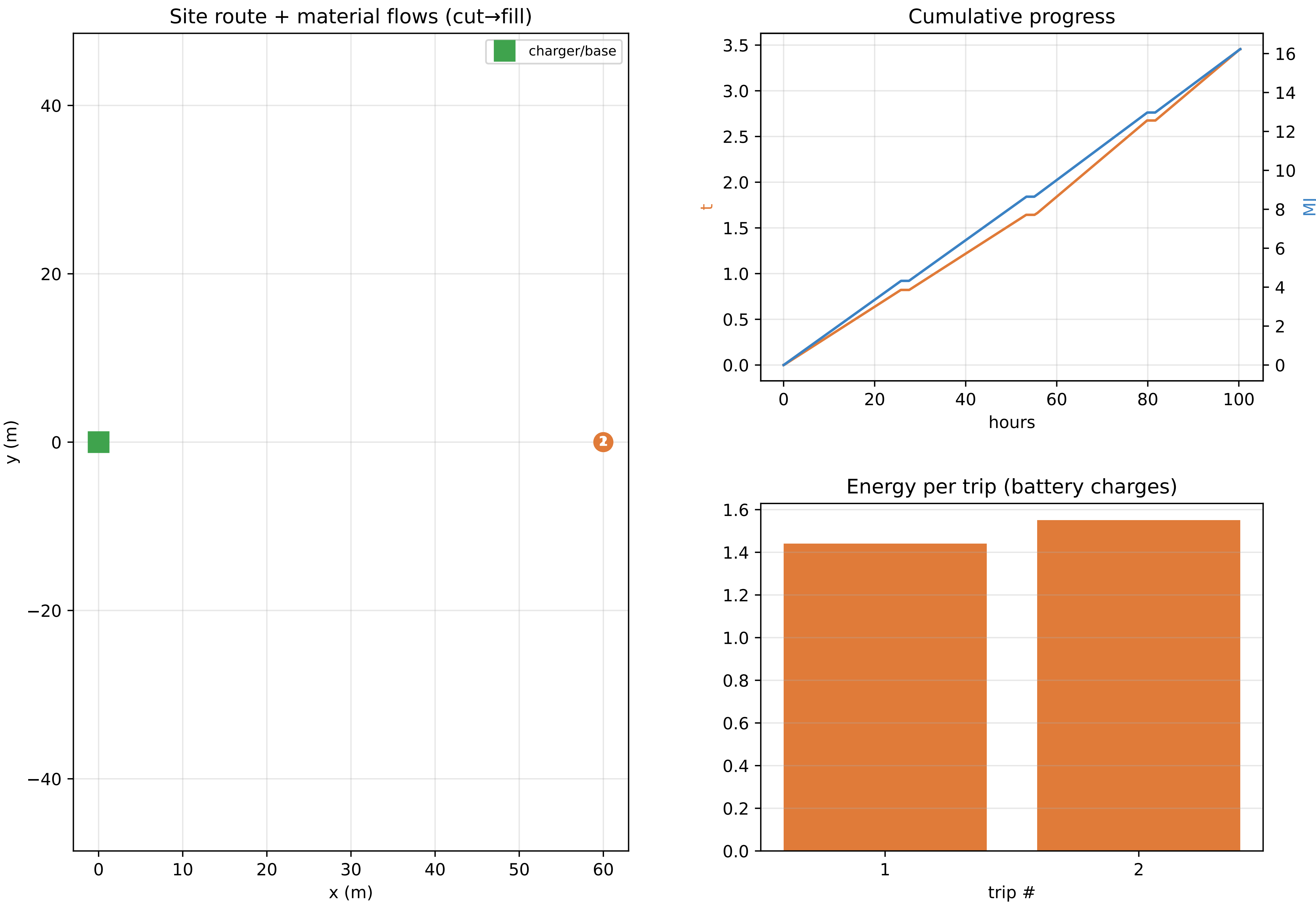
Cut-fill balanced (excavated material routed to nearest fill; surplus→spoil, deficit→import). Grounded: per-body density/gravity (bodies.json); IPEx — 0.30 m/s, 42 kg/hr dig, 4151 J/kg, 135 J/m (slip-adjusted on a DEM), 4.79 MJ battery, 30 kg/drum; SINTER 0.92 MJ/kg [CALIB] (~220× dig). Recharge 700 W + sinter-head 1000 W are [CALIB]. Pluggable sequencer × objective; battery-aware mid-task recharge.


Battery draw over the planned project
battery %
dig
drive
charge
sinter
0
20
40
60
80
100
Speed profile
speed m/s
0.00
0.05
0.10
0.15
0.20
0.25
0.30
mission time (hours)


Site route + material flows (cut→fill)
charger/base
y (m)
x (m)
Cumulative progress
t
MI
hours
Energy per trip (battery charges)
trip #